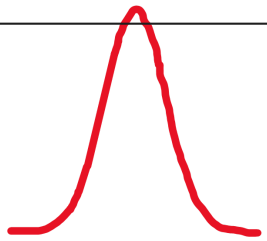
17 - Confidence interval 2

Given population distribution with mean μ, the sampling distribution of size n looks like:

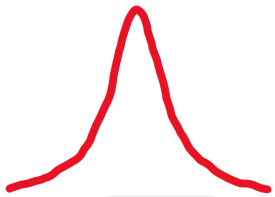
For σ known:



Sampling distribution of $\bar{x}$ is normal with _____ and _____

Sampling distribution of _____ is normal with _____ and _____

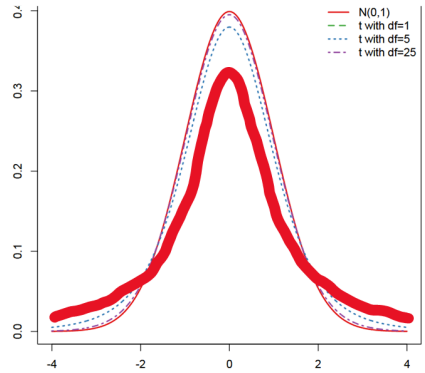
For σ unknown:



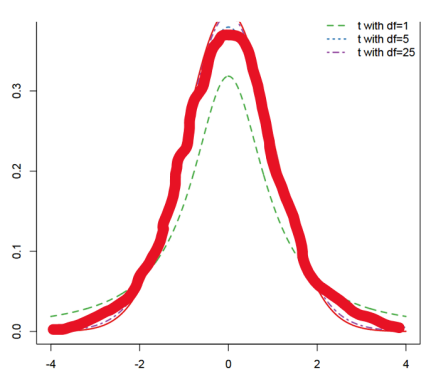
Sampling distribution of _____ is the _____-distribution.

The _____ is the _____ of the _____-distribution.

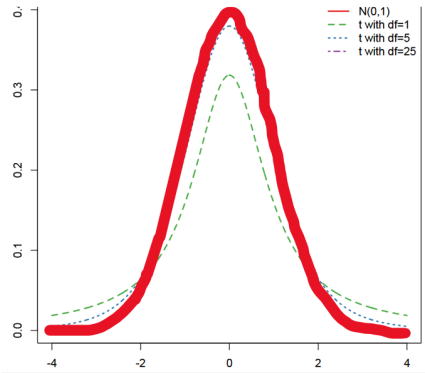
Student's _____-distribution is a family of symmetric curves _____ than the normal curve. As sample size n increases, it becomes more normal.



_____-distribution



_____-distribution



_____-distribution

Facts about Confidence Interval of Mean with σ known

The c-confidence interval for population mean μ obtained from an SRS with sample mean $\bar{x}$ and sample sd s_x is

with margin of error

when the following conditions are met:
10% Condition: $N \geq 10n$.
Approx. normal: Pop. distribution is normal or $n \geq 30$.

Here	(R command)

Example 1. Construct a 95% confidence interval for mean NOX exhaust level in all US cars given a sample of 40 readings with mean 1.268 and sample sd 0.333.